Coarse Marginal Resemblance and Model-Dependent Treatment-Response Uncertainty in a Fixed LLM Persona Panel

Yohei Nakajima
Untapped Capital, Bellevue, Washington, USA
Correspondence: yohei@untapped.vc
ORCID: https://orcid.org/0009-0009-1293-6053

Abstract

Large language models are used as synthetic participants, often evaluated through resemblance to human marginals. In sixteen fixed GPT-4.1 persona prompts, both high-continuation condition means fell inside an earlier bare-instrument reference band; extending that band to personas was exploratory. Between-prompt composition was substantial but prior-sensitive. Represented-continuation contrasts were +0.083 and +0.078. Pooled component-binomial working-model projections included zero; fixed-panel Dirichlet credible intervals excluded zero under α=.10/.25 priors but included it under Jeffreys and uniform priors. A paired-prompt bootstrap excluded zero for one wording family only and targeted a hypothetical generator. Process and text changed jointly; mechanisms remain confounded. Author-commissioned AI review prompted corrections without rewriting sealed evidence. Results establish neither human treatment-effect error nor human substitutability. Marginal resemblance and response uncertainty require separate targets and assumptions.

1. Introduction

A synthetic participant may resemble a human aggregate without reproducing the response object an experiment needs. We examine one fixed sixteen-prompt panel in strategic interaction. An exploratory comparison put its high-continuation means within a historical human-reference band, but response uncertainty depended strongly on the inferential target, prior and method. This article corrects an earlier conflation of exploratory persona comparisons with a registered bare-instrument checkpoint and reports all new saved-data contrast sensitivities, including positive intervals. This question belongs to experimental validation rather than a general verdict on artificial intelligence. Algorithmic fidelity (Argyle et al., 2023), simulated economic agents (Horton, 2023), and behavioral similarity tests (Mei et al., 2024) provide reasons to investigate synthetic participants. Li and Ji (2026) show that statistical realism can diverge from treatment-effect accuracy; Persson et al. (2026) state causal-surrogacy assumptions; Lin et al. (2026) diagnose latent-user drift under intervention. We do not claim to discover that divergence or exhaust its mechanisms.

The closest JESA comparison is Polachek et al. (2025), who compare LLM prisoner's-dilemma choices with a human experiment while varying stakes, order, and framing. Their stakes comparative-static comparison already addresses behavioral response as well as framing. Our narrower addition is a fixed-panel decomposition within a registered experimental program of between-prompt composition, within-prompt variation, and represented continuation response, coupled with an explicit dependence and correction record. Unlike their human comparison, our protocol-nonmatched human references cannot identify human treatment-effect error.

Repeated-game prompting and parameter variation also precede this study (Akata et al., 2025; Pal et al., 2026). Strong effect forecasting across human studies is positive evidence for a different use: Ashokkumar et al. (2026) forecast experimental effects, rather than construct the subject-level response surface examined

here. Calibration (Hullman et al., 2026) and latent-drift diagnostics can complement our fixed-panel analysis. The later respondent-coverage study of Kumagae and Kaneko (2026), posted after our August preprint, likewise distinguishes answer marginals from joint respondent profiles; its survey coverage target differs from our strategic-response target. Supplement S1 retains the broader literature and exact versions.

2. Design and estimands

Requests named `gpt-4.1`; captured responses reported `gpt-4.1-2025-04-14`. The alias was not an immutable request pin. The deployment used temperature 0.7, and outputs capped at sixteen tokens. A minimal subject prompt supplied no reasoning scaffold or game-theory vocabulary. The OpenAI-compatible adapter supplied temperature and `max_tokens=16`; `top_p=1` was specified in the assembly but omitted from the wire. Other sampling penalties inherited provider defaults. No tools or native structured output were used. Environment randomness was seeded; provider generation was not claimed to be seeded.

Before Phase 5 dispatch, a seeded registry generated sixteen complete persona sentences from two age bands and three binary trait dimensions: agreeable/competitive, patient/impulsive, and risk-averse/risk-seeking. Cooperative leaning meant at least two of agreeable, patient, and risk-averse, yielding eight prompts per stratum without reference to behavior. Names and occupations are uncontrolled semantic components. Each complete sentence was prepended to sealed task text. The panel is a fixed construction, not a probability sample of people or a persona generator.

Phase 5 crossed these prompts with four repeated prisoner's-dilemma cells (continuation probabilities .10/.90 × wording families S2 absent/present), a one-shot label swap, and a Community frame. Primary repeated cells have six complete episodes each. Two seats share an episode; calls, rounds, and seats are not independent subjects. For prompt i, condition d, and episode e, the round-one cooperation outcome is $Y_{ide} \in \{0, 0.5, 1\}$, the fraction of its two seats cooperating. Define $p_i(d) = E[Y_{ide} \mid i, d]$ and the equally weighted fixed-panel mean $\bar{p}(d) = 16^{-1} \sum_i p_i(d)$. The aggregate represented-treatment contrast is $\Delta = \bar{p}(.90) - \bar{p}(.10)$ within each wording family.

The continuation intervention changed the stochastic environment and the sentence communicating it jointly. It identifies response to that represented treatment, not an isolated incentive effect. Reusing explicit prompt text identifies a prompt-indexed contrast; it does not establish a stable latent person's potential outcomes. The deployment, prompt, condition, episode, seat, round, and request levels must remain distinct.

Phase 3 tested the original bare J/F instrument; Phase 4 studied representation, labels/payoffs, continuation, adversaries and monitoring; Phase 5 added personas. Phase 1–2 development is exploratory. Registration status rests on phase-specific flags, seals and chronology (Supplement S2), with incomplete external timing evidence for Phase 3. X1 was sequentially registered after Phase 3. Result-informed prospective amendments and post-adjudication sensitivities are separate tiers.

Protocol glossary. "Bare" means no persona prefix; the original J/F instrument differs from the selected D1 one-shot presentation used in later repeated assays. S2 absent/present replaces and repositions the continuation sentence jointly. P5-1a tests whether the restricted interior-cell fraction is below .10; P5-1b compares dispersion with a .75× nonmatched-human SD heuristic. P5-2 calls the pooled task-consistent share persona-dominant if its historical upper bound is ≤.20. P5-3(a) searches for any prompt×wording pair with both condition intervals inside (.05,.95) and a positive slope lower bound; its frozen search lacked family control.

2.1 Uncertainty and validation criteria

We use the episode, retaining uncertainty when observed cells are unanimous. Write $A = \mathbf{1}(Y \geq 0.5)$ and $B = \mathbf{1}(Y = 1)$, so $Y = (A + B)/2$. Simultaneous Clopper–Pearson component intervals project onto condition means. Four component-condition intervals split the error rate by Bonferroni for each contrast. If the projected intervals are $[L_d, U_d]$, the reported contrast interval is $[L_{.90} - U_{.10}, U_{.90} - L_{.10}]$. Each binary component receives a 98.75% interval; each projected condition mean has at least 97.5% coverage and the contrast at least 95%, under independent episodes and a pooled component-binomial working model. Simultaneity covers four components within one contrast, neither both contrasts nor all prompt rows; prompt heterogeneity and the precise coverage domain require care (Supplement S3). It is distinct from treating two seats as independent Bernoulli trials (Clopper and Pearson, 1934).

Three composition views are intentionally separated. Finite-opportunity-corrected plug-in estimates and conditional episode bootstraps describe the recorded panel; unanimous cells remain degenerate in that bootstrap. Symmetric Dirichlet priors on each prompt-cell's three outcome probabilities propagate sparse-cell uncertainty but are prior-sensitive. A two-stage prompt-plus-episode bootstrap changes the target toward a hypothetical persona generator. These are different conditioning assumptions, not independent replications of one estimate.

Table 1 distinguishes descriptive resemblance from registered checkpoints and inferential targets. P3-A3 registered [.36,.63] only for pooled bare high-continuation (.75/.90) means; the Phase 5 seals contain no persona extension of that band. The historical persona scorecard is exploratory. Broad condition-specific bands $[\ell_d, u_d]$ imply only $\Delta \in [\ell_1 - u_0, u_1 - \ell_0]$. Exact matching of both means would fix their difference; our point concerns slack in coarse criteria. The law of total variance and classical coupling results explain why marginal moments do not determine between/within composition or cross-condition coupling (Supplement S3); these applications are not new probability theorems.

Target	Criterion or estimate	What it supports
Marginal level	Both high-continuation persona means inside earlier [.36,.63] band	Exploratory resemblance, no registered persona-band pass
Composition	Plug-in 85–96%; Dirichlet medians depend on prior	Substantial prompt-indexed variation in this panel
Aggregate response	+.083 / +.078; intervals differ by method/prior	Model-dependent represented-response uncertainty
Latent-person response	No invariance test	Unidentified
Human causal accuracy	Protocol-nonmatched references	Unidentified

Table 1. Validation targets, evidence, and inference limits. Composition percentages are between-prompt shares of episode-level variation. Neither the bands nor the variance heuristic is an equivalence margin.

3. Results

3.1 Marginal fit and composition

Repeated-cell cooperation means were 0.349 and 0.427 at continuation .10, and 0.432 and 0.505 at .90. Both high-continuation means fall inside P3-A3's earlier bare-instrument [.36,.63] band. Counting three of

all four means inside that band is an exploratory extension, not a registered Phase 5 validation result. The original low-continuation human anchors were .190 and .1783; both persona low-condition means exceed them. These protocol-nonmatched comparisons do not identify human response error. Cooperative-leaning minus defect-leaning prompt-bundle differences were 0.510–0.719. They are not isolated causal trait effects.

Between-prompt plug-in shares were 85.5%, 96.1%, 88.8%, and 90.2%. Fixed-panel Jeffreys Dirichlet$(0.5, 0.5, 0.5)$ posterior medians were 63.1%, 70.5%, 66.1%, and 66.6%; respective 95% intervals were [49.4%,74.5%], [57.3%,81.3%], [52.7%,77.1%], and [52.7%,77.7%]. A symmetric-prior sweep yielded medians 74.8–82.5% at $\alpha = .25$ and 47.1–53.5% at $\alpha = 1$. Thus substantial composition persists, but a share exceeding one-half is prior-dependent (Figure 1).

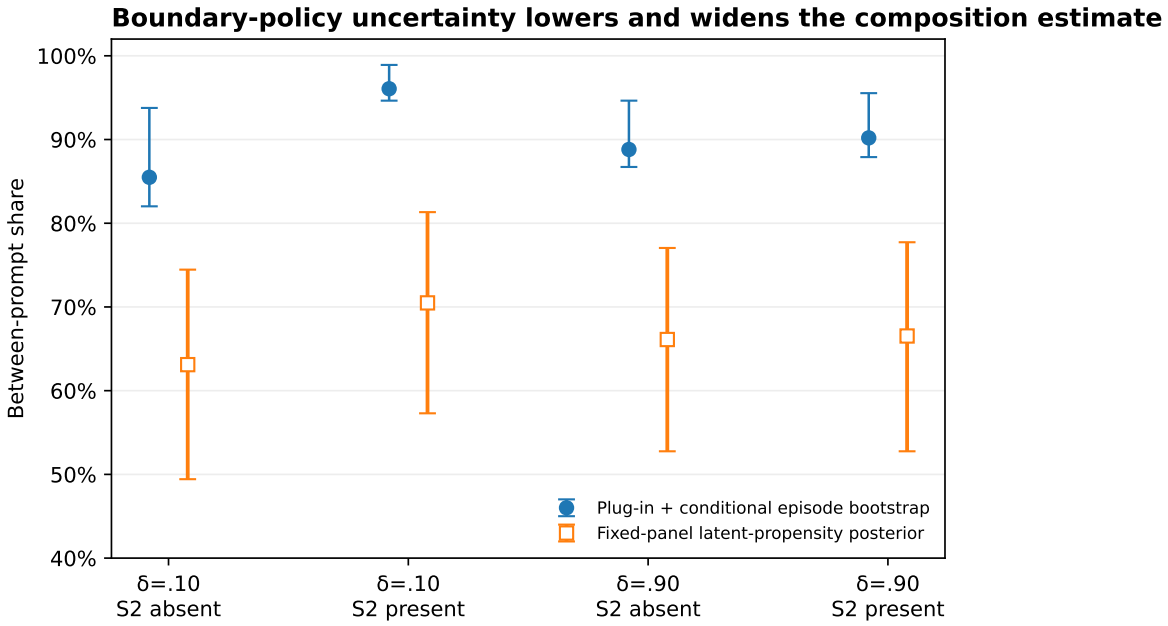


Figure 1. Plug-in/conditional-bootstrap and Jeffreys fixed-panel composition estimates. The former conditions on recorded corner concentration; the latter shrinks sparse cells inward. Uniform-prior medians reported in text straddle one-half. These estimates are not interchangeable.

P5-1b compared corrected dispersion with heuristic thresholds derived from Dal Bó and Fréchette's (2011) human strategy frequencies. Those experiments differ in continuation, payoffs, incentives, assignment, and experience. The frozen ratio .75 is a tolerance, not a human-equivalence margin. The historical dispersion checkpoint is permissive; neither its passage nor matched-looking SD establishes human variance equivalence.

The binary interiority census is also method-sensitive. P5-1a supported its corner-mixture predicate when the restricted interior fraction was below .10. The frozen seat count 3/32=.094 and post-adjudication episode-projection count 2/32=.063 satisfy that threshold; the Jeffreys count 5/32=.156 does not. This unfavorable sensitivity reverses the predicate while leaving its historical verdict unchanged. Across all six conditions, the corresponding counts are 14/96, 11/96, and 19/96. Six episodes and discrete outcomes make threshold labels unstable; continuous estimates are more informative.

3.2 Response and representation

Continuation contrasts were +0.083 (S2 absent) and +0.078 (S2 present). Under the pooled component-binomial working model, simultaneous 95% intervals were [−0.171,+0.330] and [−0.181,+0.330]. The observed contrasts are descriptive arithmetic; these wide working-model projections are not the uniquely appropriate uncertainty analysis for a heterogeneous fixed panel (Figure 2). Table 2 reports the additional post-adjudication sensitivities. The α=.10/.25 fixed-panel credible intervals exclude zero for both contrasts; Jeffreys and uniform intervals include it. The paired-prompt bootstrap excludes zero only for S2 absent. These are distinct assumptions and targets, not independent votes for or against an effect.

Method / target	S2 absent interval	S2 present interval
Pooled working-model CP / fixed-panel means	[−.171,.330]	[−.181,.330]
Dirichlet α=.10 / fixed prompts	[.0285,.1293]	[.0299,.1183]
Dirichlet α=.25 / fixed prompts	[.0140,.1329]	[.0135,.1245]
Dirichlet α=.50 / fixed prompts	[−.0023,.1340]	[−.0038,.1270]
Dirichlet α=1 / fixed prompts	[−.0188,.1292]	[−.0204,.1242]
Paired-prompt two-stage bootstrap / hypothetical generator	[.0104,.1771]	[−.0104,.1875]

Table 2. CP intervals have per-contrast simultaneous 95% coverage only under the stated working model. Dirichlet intervals are marginal central 95% credible intervals under independent prompt-condition priors over {0,.5,1}; the bootstrap intervals are marginal central 95% percentiles, without design-based generator coverage. Neither has a simultaneous two-contrast guarantee. Settings were declared before this saved-count computation, after model review: 50,000 draws per α and 50,000 bootstrap draws, all reported (Supplement S3.1). They are not preregistered confirmation.

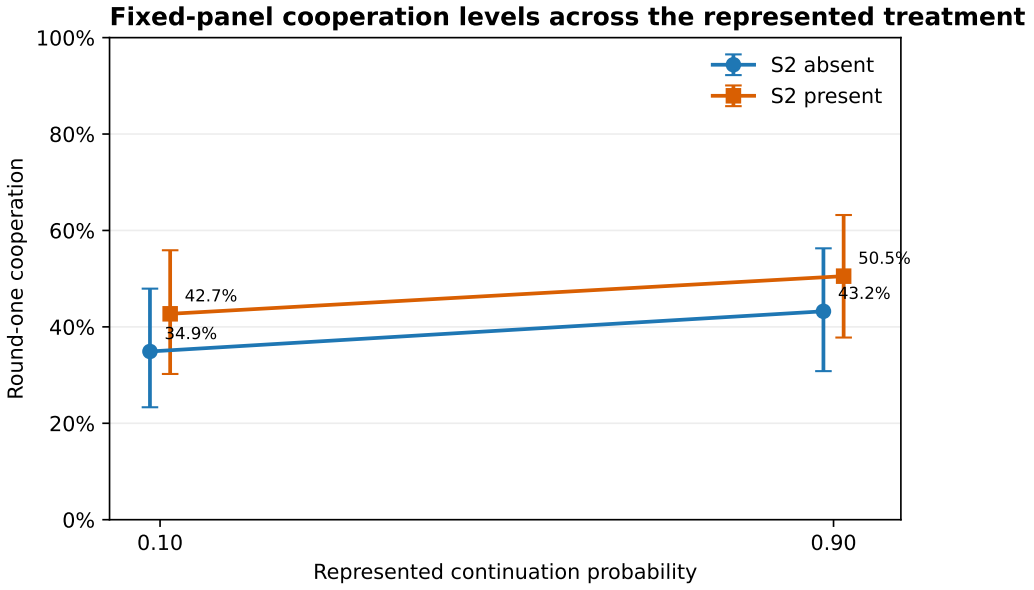


Figure 2. Cooperation levels with ≥97.5% projected condition intervals under the pooled component-binomial working model. Each contrast uses four 98.75% component intervals for its own ≥95% coverage; there is no joint guarantee across both contrasts. Lines orient the eye; contrast intervals in text use simultaneous component projection. Process and communicating text change together.

For the bare configuration, a screened and held-out S2 wording-and-position replacement shifted cooperation from 0/40 to 37/40 seat choices. Wording and position were one operation. By contrast, ordinary one-shot wording had a registered main effect of +0.0063 (SE .0210; Holm-adjusted $p = 1.00$). Announcing and implementing repeated interaction produced larger changes, including ceiling-confounded cells; those ceilings cannot identify a continuation slope.

With canonical payoffs held fixed, swapping "Cooperate" and "Defect" made the bare configuration always choose the dominated role labeled "Defect." This demonstrates label-linked control in that cell, but learned game associations and intrinsic lexical valence are not separated. Adding a persona reversed the bare choice; no format-matched neutral prefix was tested, and label and payoff then favored the same option. Neither operation isolates a universal semantic mechanism.

P5-2 pooled 90/704 seat decisions across 352 episodes, equivalently an episode mean of 45/352=.128. A generator-oriented stratified bootstrap over forty prompt-by-conflict clusters gave [.071,.189]. Fixed-panel Dirichlet aggregation moved from median .138 at $\alpha = .10$ to .205 at $\alpha = 1$, whose interval [.182,.231] crosses the registered .20 boundary. The historical verdict remains visible; Bayesian proximity to its boundary is not prior-robust corroboration. Repeated conflict subcells were mixed; the pooled dominance classification is carried by the word/payoff-confounded swap cell.

3.3 Multiplicity and the withdrawn persona interpretation

The frozen P5-3(a) search selected persona p13 after searching multiple persona-by-wording candidates without prospective family control. Its apparent capability finding is withdrawn. Three post-adjudication 200,000-permutation analyses dynamically reapply their eligibility gates in every draw. Add-one family-wise p values and Monte Carlo 95% intervals are .059230 [.058194,.060268] under the historical seat gate; .043455 [.042561,.044353] under the percentile episode-bootstrap gate; and .773206 [.771363,.775039] for the largest eligible candidate p05 under the component-binomial working-model gate, where p13 is ineligible.

The favorable bootstrap variant cannot create prospective confirmation; none was sealed before computation. Conversely, the working-model gate cannot decisively disconfirm the capability. At six episodes, its eligible compositions permit a maximum slope .333; the archived family's estimated minimum attainable familywise tail probability is .075040 [.073884,.076198], above .05. P13 is a prospective replication target, not an established finding or a decisive negative. No new study is needed for the present bounded conclusion.

No study-wide alpha allocation was registered across P5 predicates and X1. Registered block-specific controls, including Holm-controlled adversary tests, retain their own scope. The complete supplementary record retains favorable and unfavorable variants, dead predictions, invalidations, and secondary findings. Mechanical predicate enforcement cannot ensure construct validity or an appropriate family.

4. Implications and limitations

Validation should name the target, intervention representation, dependence unit and uncertainty assumptions. Here the marginal resemblance comparison was exploratory, and response evidence changed with prior and resampling target. Positive smaller-α intervals must not be suppressed, while zero-containing intervals cannot establish zero or equivalence. No protocol-matched human comparison identifies treatment-effect error. The study demonstrates model-dependent interpretation in one fixed deployment, rather than a general failure of synthetic response.

The model version and sixteen bundled prompts bound generalization. Prompt pairing does not establish latent-person invariance; no new human comparison, wider persona sample, factorial mechanism control, or endpoint refresh is claimed. Gemini observations are descriptive because its unversioned endpoint drifted. Temperature interactions outside the frozen plan are exploratory. More episodes may improve within-prompt precision, but a wider persona-generator claim requires a declared sampling design.

Auditability enabled correction rather than self-validation. A sealed pre-data discussion originally interpreted p13 as an incentive-response existence result; author-commissioned AI review exposed dependence, family-error, and construct defects. The original discussion, mechanical verdicts, seals, and amendment history remain unchanged. One AI reviewer later helped specify post-adjudication analyses; this role transition and later reviewer-supplied reconstruction checks (tool identities unrecorded) are disclosed in Supplement S2. The human author retains responsibility. These records preserve the distinction between registered predictions, result-informed prospective changes, descriptive observations, and later reinterpretation.

5. Data, code, and declarations

The immutable study baseline is repository commit 62ab90beff1d291b069c87f13b6b6b3495171ddd; the distinct public preprint is arXiv:2608.00979v1. The separate preparation evidence ancestor 1bc9cb8dcb44cf514fe2fad53cdbfd96725badd0 contains the offline wrapper absent from the study baseline and the new contrast plan/script/complete results: https://github.com/yoheinakajima/synthetic-players/tree/1bc9cb8dcb44cf514fe2fad53cdbfd96725badd0/journal/jesa. The final editorial commit is recorded separately by the candidate manifest/PR. Supplement S4 and Data and Instructions distinguish checkout routes and link actual license text (`LICENSE`), code, registrations, prompts, dictionary and receipts. The archive contains 4,916 confirmatory runs: 320 Phase 3/X1 LLM, twenty deterministic baselines, 2,864 Phase 4, and 1,712 Phase 5. Three legacy diagnostics raise the verifier boundary to 4,919 runs, of which 4,899 are LLM runs; runs are not provider calls.

Phase 3 lacks the richer response IDs and request-body hashes of Phases 4–5. The latter contain 30,421 request events and 30,397 response events; twenty-four failed-run partials are excluded from completed analyses but retained in accounting. Snapshot checksums and timestamps detect changes relative to publication; they are not provider attestation or proof of immutability from receipt. Replay reconstructs archived behavior without live models and cannot establish human validity or adequate power.

Funding and competing interests. This research, including computational resources and model/API usage, was supported by Untapped Capital. The author is its Founder and General Partner and has a financial interest through that role. Untapped Capital lists Explore Science as a portfolio company; the author used Explore Science for manuscript review. The author is also a developer of ActiveGraph, the event-sourced runtime used here. Support was through the author and firm, rather than an independent third-party sponsor.

AI assistance and author responsibility. Subject-model observations (GPT-4.1, descriptive Gemini, and a failed Claude Haiku entry candidate) are distinct from assistance with research. Author-commissioned Claude and Explore Science AI reviews informed critique and revision; Replit Agent contributed implementation, analysis integration, and manuscript revision. The Round 2 methods reviewer later helped specify analyses. Exact underlying assistance-model versions are not recorded. Codex, with task configuration gpt-6.1-sol/high, assisted the November 2026 journal compression, citation audit, offline verification wrapper, and source/PDF preparation; this configuration is not provider attestation. Supplement S2.1 records sources, dates, roles, and unknowns. These author-side AI reviews are not journal peer review. AI systems are not authors. The sole human author selected the questions, approved registered designs, adjudicated review recommendations, and accepts responsibility without claiming independent rederivation of every result.

References

Akata, E., Schulz, L., Coda-Forno, J., Oh, S. J., Bethge, M., and Schulz, E. (2025). Playing repeated games with large language models. *Nature Human Behaviour, 9*, 1380–1390. https://doi.org/10.1038/s41562-025-02172-y

Argyle, L. P., Busby, E. C., Fulda, N., Gubler, J. R., Rytting, C., and Wingate, D. (2023). Out of one, many: Using language models to simulate human samples. *Political Analysis, 31*(3), 337–351. https://doi.org/10.1017/pan.2023.2

Ashokkumar, A., Hewitt, L., Ghezae, I., and Willer, R. (2026). Large language models can predict the results of social science experiments. *Nature, 656*, 115–122. https://doi.org/10.1038/s41586-026-10742-x

Clopper, C. J., and Pearson, E. S. (1934). The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika, 26*(4), 404–413. https://doi.org/10.1093/biomet/26.4.404

Dal Bó, P., and Fréchette, G. R. (2011). The evolution of cooperation in infinitely repeated games: Experimental evidence. *American Economic Review, 101*(1), 411–429. https://doi.org/10.1257/aer.101.1.411

Horton, J. J. (2023). Large language models as simulated economic agents: What can we learn from Homo Silicus? NBER Working Paper 31122, original April 2023 version. https://www.nber.org/system/files/working_papers/w31122/revisions/w31122.rev0.pdf. https://doi.org/10.3386/w31122

Hullman, J., Broska, D., Sun, H., and Shaw, A. (2026). This human study did not involve human subjects: Validating LLM simulations as behavioral evidence. arXiv:2602.15785v1. https://arxiv.org/abs/2602.15785v1. https://doi.org/10.48550/arXiv.2602.15785

Kumagae, Y., and Kaneko, M. (2026). Silicon samples match the answers but not the respondents. arXiv:2610.30900v1. https://arxiv.org/abs/2610.30900v1. https://doi.org/10.48550/arXiv.2610.30900

Li, Z., and Ji, F. (2026). Statistical realism is not evidence that LLMs can estimate treatment effects in social science experiments. arXiv:2604.02458v3. https://arxiv.org/abs/2604.02458v3. https://doi.org/10.48550/arXiv.2604.02458

Lin, V., Yun, T., Matarić, M. J., Canny, J., Gretton, A., and D'Amour, A. (2026). The illusion of intervention: Your LLM-simulated experiment is an observational study. arXiv:2605.20767v1. https://arxiv.org/abs/2605.20767v1. https://doi.org/10.48550/arXiv.2605.20767

Mei, Q., Xie, Y., Yuan, W., and Jackson, M. O. (2024). A Turing test of whether AI chatbots are behaviorally similar to humans. *Proceedings of the National Academy of Sciences, 121*(9), e2313925121. https://doi.org/10.1073/pnas.2313925121

Pal, S., Mallela, A., Hilbe, C., Pracher, L., Wei, C., Fu, F., Schnell, S., and Nowak, M. A. (2026). Strategies of cooperation and defection in five large language models. arXiv:2601.09849v1. https://arxiv.org/abs/2601.09849v1. https://doi.org/10.48550/arXiv.2601.09849

Persson, J., Schultzberg, M., and Ankargren, S. (2026). Statistical foundations of LLM-based A/B testing: A surrogacy framework for human causal inference. arXiv:2606.17165v3. https://arxiv.org/abs/2606.17165v3. https://doi.org/10.48550/arXiv.2606.17165

Polachek, S. W., Romano, K., and Tonguc, O. (2025). Homo-silicus: not (yet) a good imitator of homo sapiens or homo economicus. *Journal of the Economic Science Association, 11*(2), 261–269. https://doi.org/10.1017/esa.2025.10023